Data Checking

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2026-10-16

Background

Students in medicine from underrepresented backgrounds (URM) such as; American Indian/Alaska Native, Hispanic, Vietnamese, Black/African American, and Native Hawaiian, have experienced bias that has negatively impacted their ability to succeed in medicine. This bias has been observed in USMLE scores, as well as clerkship scores previously. This study aims to determine if pre-clinical Objective Structured Clinical Skills Examination (OSCE) scores for students who are URM are different from students who don't fall under that category. This study will focus on physical exam and communication exam OSCE scores for these students.

Data Checking

The following data from the University of the Colorado School of Medicine (CUSOM) for the graduating class of 2020, has issues with NMAR (not missing at random) values. When the data were examined to detect patterns in the missing data, it was first observed that the AAC_clinical_grade and the RCC_clinical_grade columns are missing the same values at the same locations. This spurred further investigation of the data with the missing OSCE scores, since they also appear to have some similar pattern to their missingness. When the data set was split into 2 separate data sets for the missing vs. non-missing OSCE data, it was observed that the individuals who were not missing their communication OSCE data were missing all of their AAC_clinical_grade and RCC_clinical_grade scores. Additionally, it was observed for individuals who were missing their communication OSCE scores, that some of them were missing some of their AAC and RCC clinical grade scores, but not all.

A reasonable explanation for this missingness is that 2 separate data sets were combined into this current data set, where some students were missing their AAC and RCC clinical grade scores, but had their OSCE scores in one of the data sets. Because the rest of the data does not appear to be skewed for or against, students with/without OSCE scores, this data can be reasonably used for statistical analysis. See Table 2 and Table 3 in the appendix for a comparison of these two data sets.

In addition, clinical grades, USMLE scores, and the averages of the physical examination and communication OSCE exams were examined and stratified by whether or not a student is underrepresented in medicine. It appears from a brief examination of Table 1, as well as the graphs presented below, that students that are underrepresented in medicine tend to do slightly worse than their peers when it comes to grading. However, further analysis is necessary to confirm if this difference is statistically significant.

Table 1: Data stratified by URM Status

Characteristic	**0** N = 132[1]	**1** N = 60[1]	**3** N = 7[1]
AAC_clinical_grade			
1	2 (17%)	3 (43%)	0 (NA%)
2	7 (58%)	1 (14%)	0 (NA%)

Table 1: Data stratified by URM Status

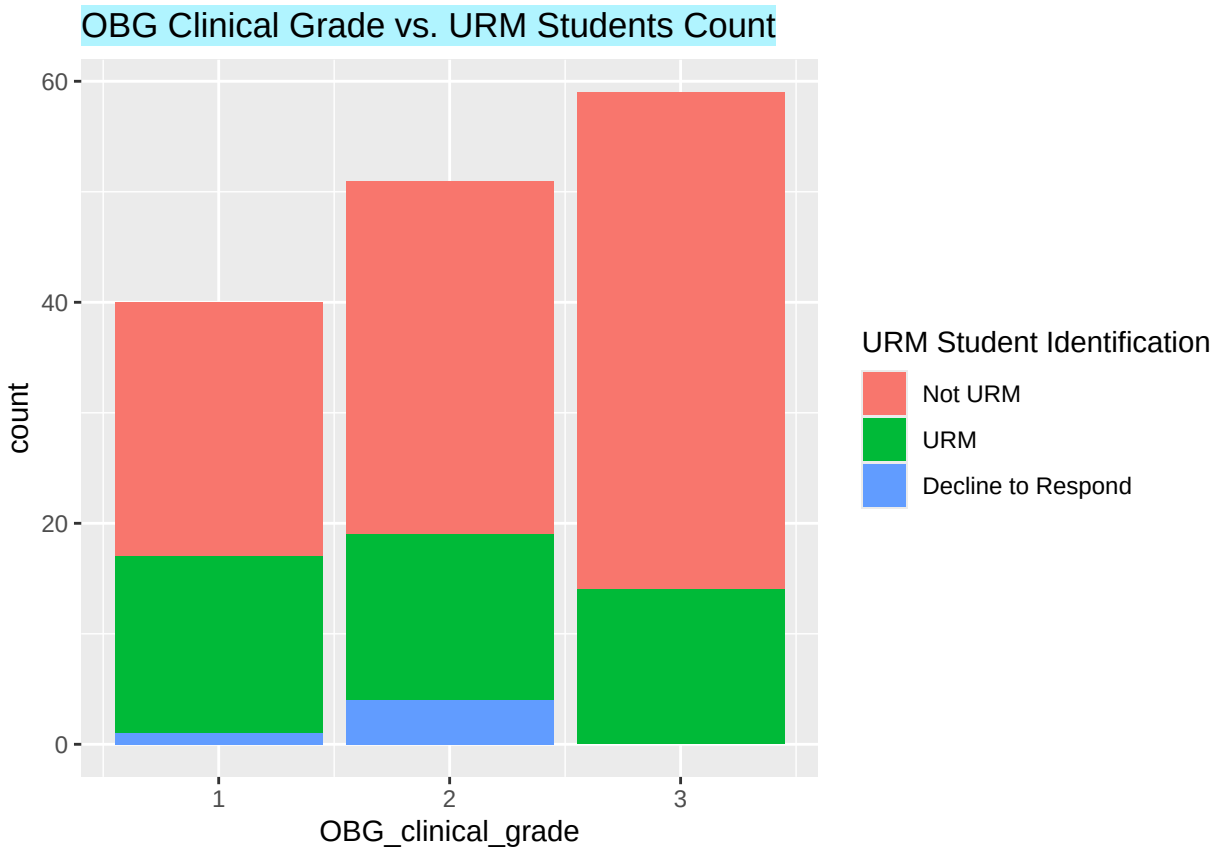
Characteristic	**0** N = 132[1]	**1** N = 60[1]	**3** N = 7[1]
3	3 (25%)	3 (43%)	0 (NA%)
Unknown	120	53	7
CPC_clinical_grade			
1	10 (11%)	4 (12%)	3 (43%)
2	31 (33%)	12 (36%)	2 (29%)
3	52 (56%)	17 (52%)	2 (29%)
Unknown	39	27	0
HAC_clinical_grade			
1	25 (21%)	16 (31%)	0 (0%)
2	35 (29%)	24 (47%)	3 (50%)
3	59 (50%)	11 (22%)	3 (50%)
Unknown	13	9	1
ICAC_clinical_grade			
1	24 (22%)	15 (27%)	1 (20%)
2	40 (36%)	25 (45%)	2 (40%)
3	47 (42%)	15 (27%)	2 (40%)
Unknown	21	5	2
NC_clinical_grade			
1	8 (7.5%)	7 (14%)	1 (17%)
2	56 (53%)	27 (55%)	4 (67%)
3	42 (40%)	15 (31%)	1 (17%)
Unknown	26	11	1
OBG_clinical_grade			
1	23 (23%)	16 (36%)	1 (20%)
2	32 (32%)	15 (33%)	4 (80%)
3	45 (45%)	14 (31%)	0 (0%)
Unknown	32	15	2
OPC_clinical_grade			
1	12 (11%)	9 (20%)	0 (0%)
2	43 (40%)	23 (52%)	4 (67%)
3	53 (49%)	12 (27%)	2 (33%)
Unknown	24	16	1
PC_clinical_grade			

Table 1: Data stratified by URM Status

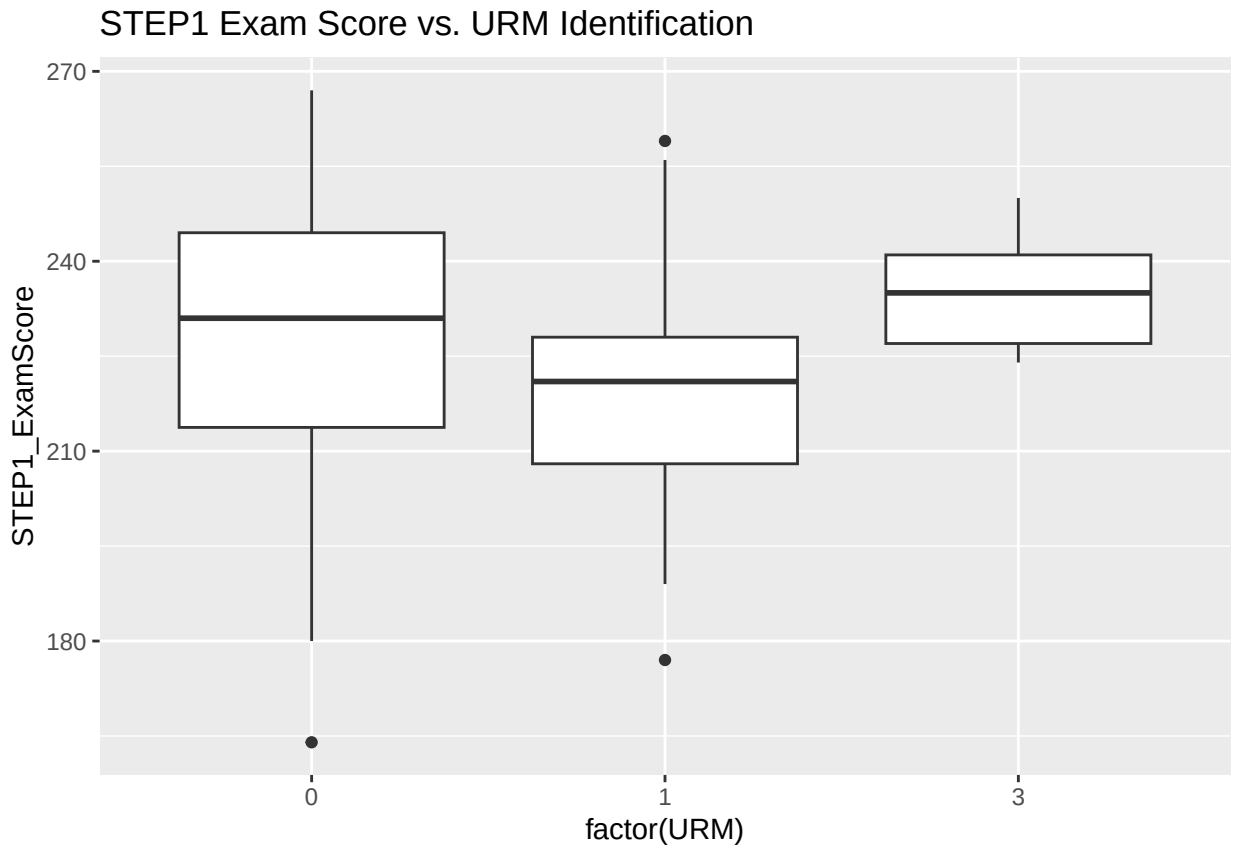
Characteristic	**0** N = 132[1]	**1** N = 60[1]	**3** N = 7[1]
1	28 (27%)	18 (42%)	3 (43%)
2	37 (36%)	12 (28%)	2 (29%)
3	37 (36%)	13 (30%)	2 (29%)
Unknown	30	17	0
RCC_clinical_grade			
1	2 (17%)	3 (43%)	0 (NA%)
2	6 (50%)	1 (14%)	0 (NA%)
3	4 (33%)	3 (43%)	0 (NA%)
Unknown	120	53	7
STEP1_ExamScore	231 (214, 245)	221 (208, 228)	235 (227, 246)
STEP2CK_ExamScore	246 (236, 256)	237 (229, 247)	247 (244, 248)
Unknown	0	1	0
gender			
D	0 (0%)	0 (0%)	1 (14%)
F	70 (53%)	29 (48%)	2 (29%)
M	62 (47%)	31 (52%)	4 (57%)
OSCE_phys_exam_avg	90.8 (88.7, 93.1)	89.7 (86.5, 92.8)	90.4 (89.1, 95.2)
Unknown	30	18	1
OSCE_comm_exam_avg	93.7 (90.1, 96.0)	92.9 (89.0, 95.9)	91.0 (89.1, 92.1)
Unknown	28	17	1

[1]n (%); Median (Q1, Q3)

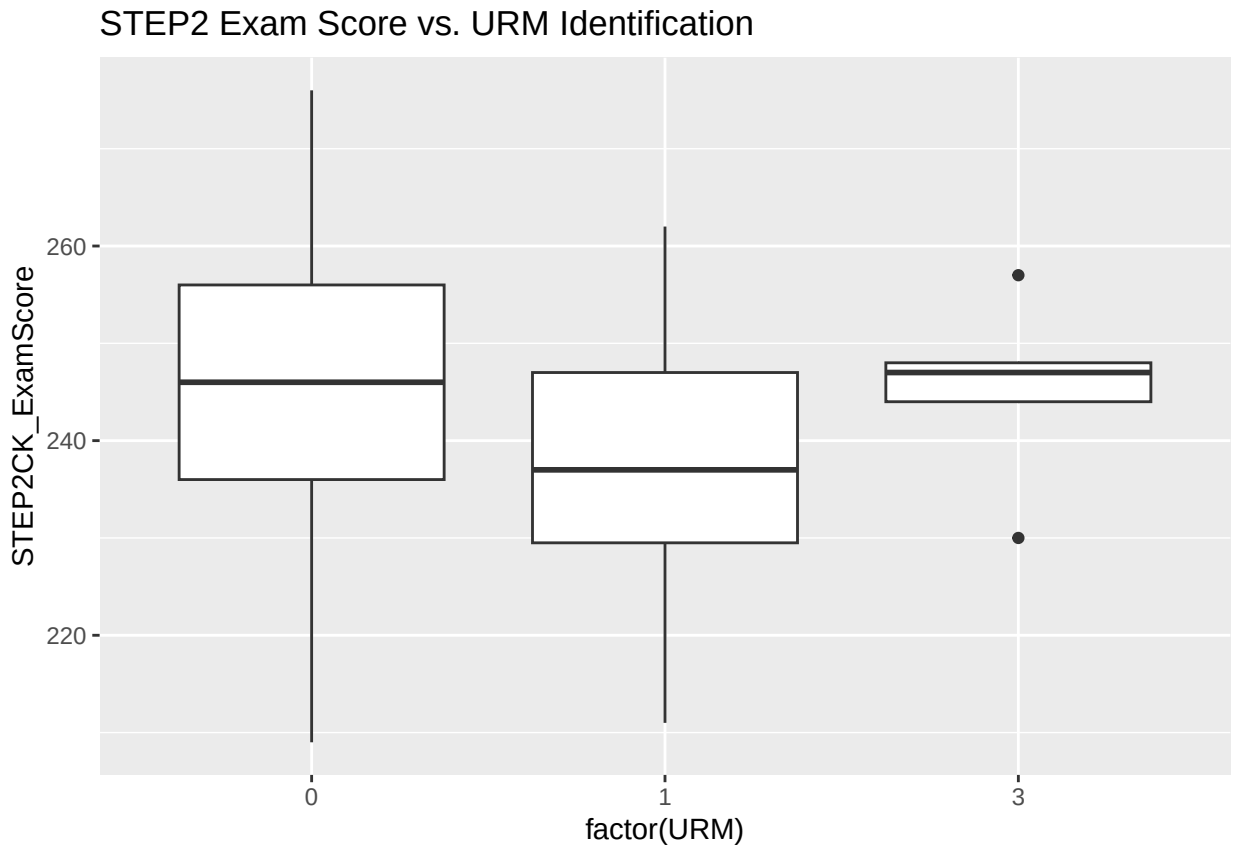
```
## Warning: Removed 49 rows containing non-finite outside the scale range
## (`stat_count()`).
```



As can be seen from the bar chart above, as well as the bar charts presented in the appendix, URM students tend to make up a smaller proportion of "honors" students, which is represented as 3 for clinical grades. In addition, they tend to make up a larger proportion of "passing" students, which is represented as 1 for clinical grades. However, from these bar charts, it is difficult to say whether these observed differences are significant between URM and non-URM students.

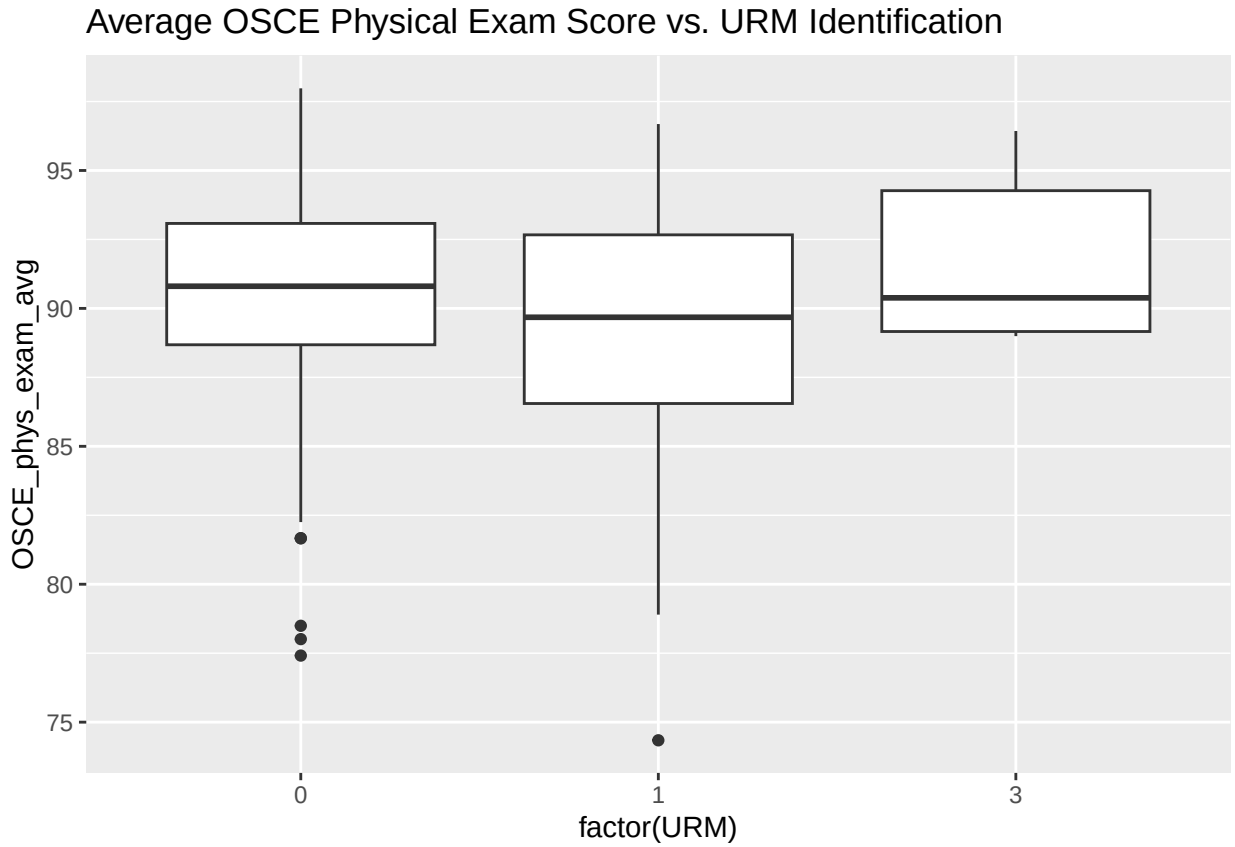



```
## Warning: Removed 1 row containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```

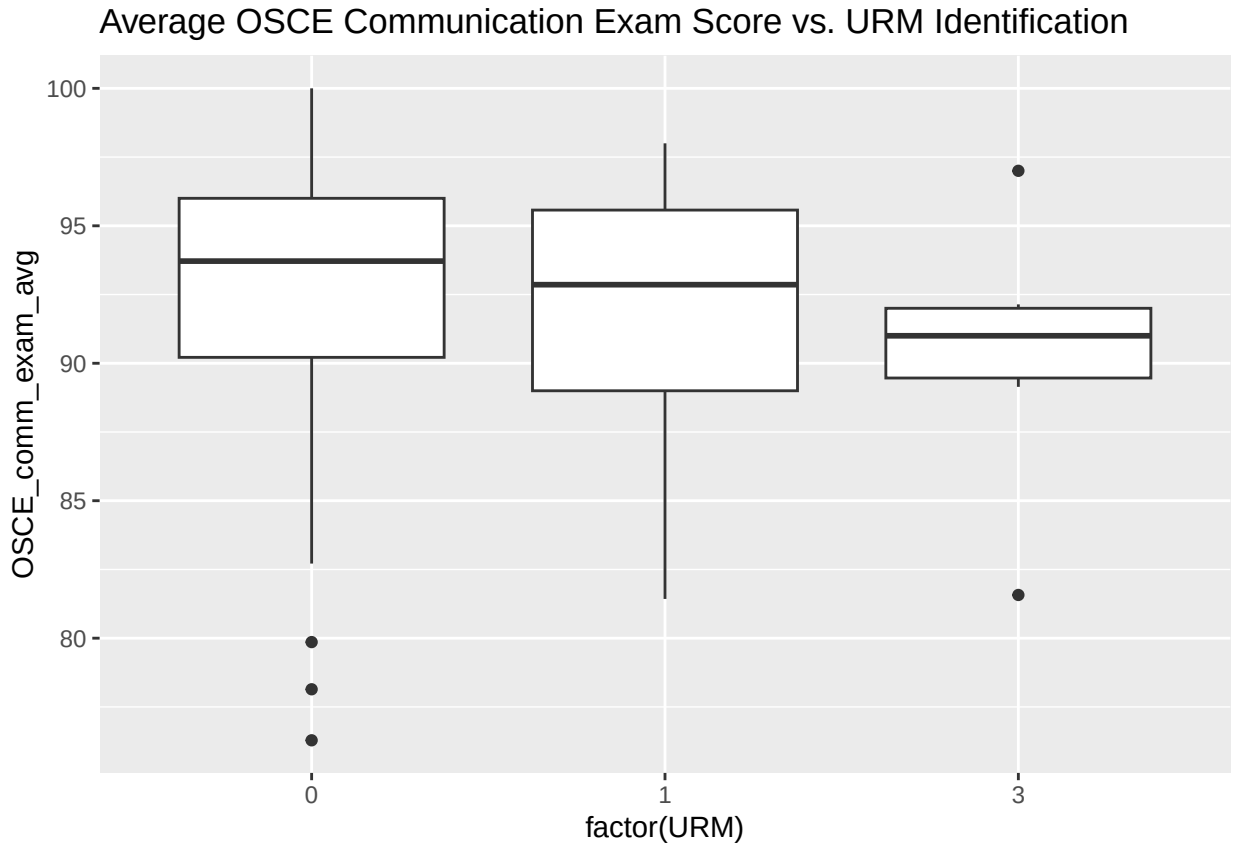


From the box plots presented above, it appears that students who identify as URM tend to have lower STEP1 and STEP2 exam scores. However, similar to above, further analysis must be conducted to determine if this is significant.

```
## Warning: Removed 49 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```




```
## Warning: Removed 46 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```



From the box plots presented above, it appears that students who identify as URM tend to have slightly lower OSCE scores. However, further analysis must be conducted to determine if this difference is significant.

Conclusion

There appears to be some irregular patterns in the data with regard to missingness, however these patterns should not impact the ability to run statistical tests on the data for the variables that Dr. Guth is interested in analyzing. In addition, there appears to be some negative bias towards URM students, however from a simple look at the data, it is difficult to tell if this difference is significant.

Figure Appendix

Table 2: Missing OSCE

Characteristic	**N = 46**[1]
AAC_clinical_grade	
1	5 (26%)
2	8 (42%)
3	6 (32%)
Unknown	27
CPC_clinical_grade	

Table 2: Missing OSCE

Characteristic	**N = 46**[1]
1	2 (10%)
2	9 (45%)
3	9 (45%)
Unknown	26
HAC_clinical_grade	
1	12 (29%)
2	16 (38%)
3	14 (33%)
Unknown	4
ICAC_clinical_grade	
1	15 (36%)
2	15 (36%)
3	12 (29%)
Unknown	4
NC_clinical_grade	
1	4 (9.8%)
2	27 (66%)
3	10 (24%)
Unknown	5
OBG_clinical_grade	
1	12 (32%)
2	8 (21%)
3	18 (47%)
Unknown	8
OPC_clinical_grade	
1	7 (18%)
2	19 (48%)
3	14 (35%)
Unknown	6
PC_clinical_grade	
1	14 (36%)
2	11 (28%)
3	14 (36%)

Table 2: Missing OSCE

Characteristic	**N = 46**[1]
Unknown	7
RCC_clinical_grade	
1	5 (26%)
2	7 (37%)
3	7 (37%)
Unknown	27
URM	
0	28 (61%)
1	17 (37%)
3	1 (2.2%)
STEP1_ExamScore	225 (216, 234)
STEP2CK_ExamScore	236 (228, 245)
gender	
F	25 (54%)
M	21 (46%)
OSCE_phys_exam_avg	NA (NA, NA)
Unknown	46
OSCE_comm_exam_avg	NA (NA, NA)
Unknown	46

[1]n (%); Median (Q1, Q3)

Table 3: **Table 3. Non-Missing OSCE**

Characteristic	**N = 153**[1]
AAC_clinical_grade	NA (NA, NA)
Unknown	153
CPC_clinical_grade	
1	15 (13%)
2	36 (32%)
3	62 (55%)
Unknown	40
HAC_clinical_grade	
1	29 (22%)
2	46 (34%)

Table 3: **Table 3. Non-Missing OSCE**

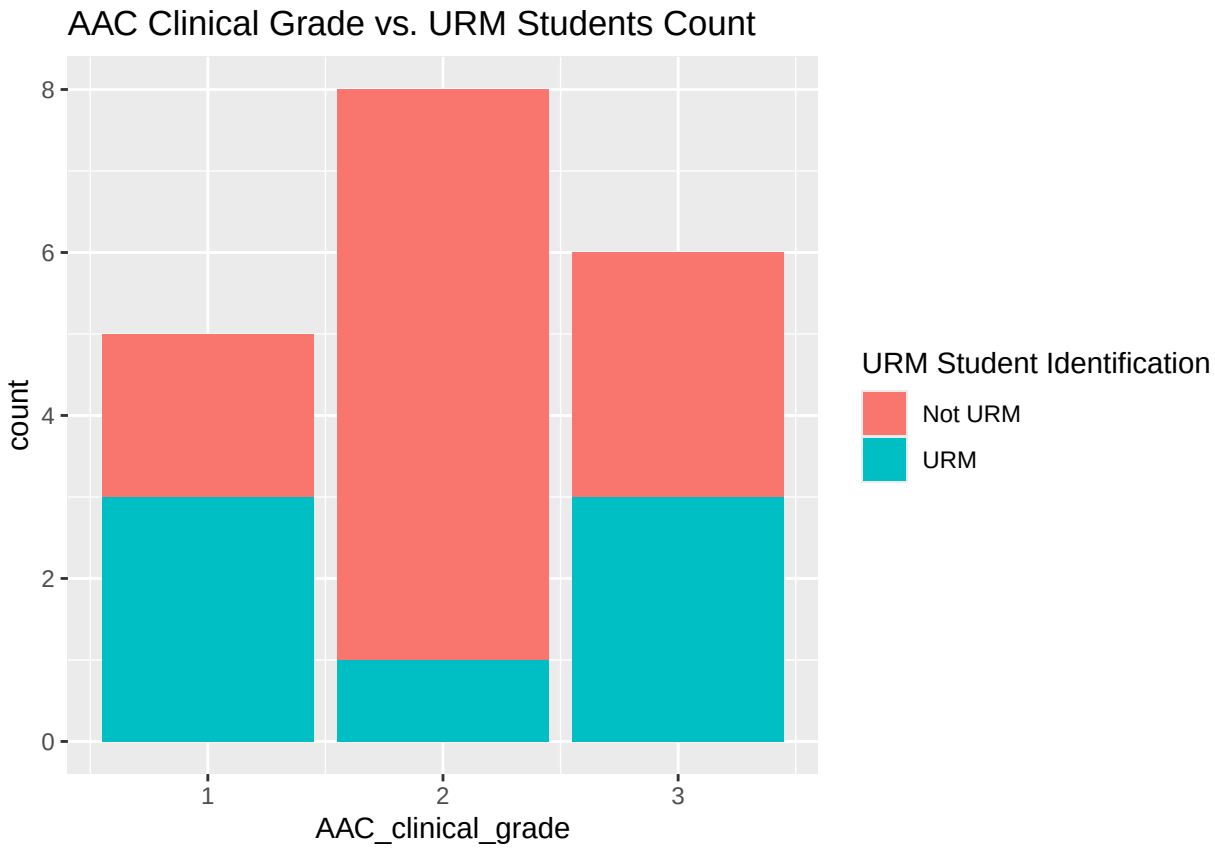
Characteristic	**N = 153**[1]
3	59 (44%)
Unknown	19
ICAC_clinical_grade	
1	25 (19%)
2	52 (40%)
3	52 (40%)
Unknown	24
NC_clinical_grade	
1	12 (10%)
2	60 (50%)
3	48 (40%)
Unknown	33
OBG_clinical_grade	
1	28 (25%)
2	43 (38%)
3	41 (37%)
Unknown	41
OPC_clinical_grade	
1	14 (12%)
2	51 (43%)
3	53 (45%)
Unknown	35
PC_clinical_grade	
1	35 (31%)
2	40 (35%)
3	38 (34%)
Unknown	40
RCC_clinical_grade	NA (NA, NA)
Unknown	153
URM	
0	104 (68%)
1	43 (28%)
3	6 (3.9%)

Table 3: **Table 3. Non-Missing OSCE**

Characteristic	**N = 153**[1]
STEP1_ExamScore	228 (211, 244)
STEP2CK_ExamScore	246 (236, 255)
Unknown	1
gender	
D	1 (0.7%)
F	76 (50%)
M	76 (50%)
OSCE_phys_exam_avg	90.6 (88.1, 93.1)
Unknown	3
OSCE_comm_exam_avg	93.6 (89.9, 96.0)

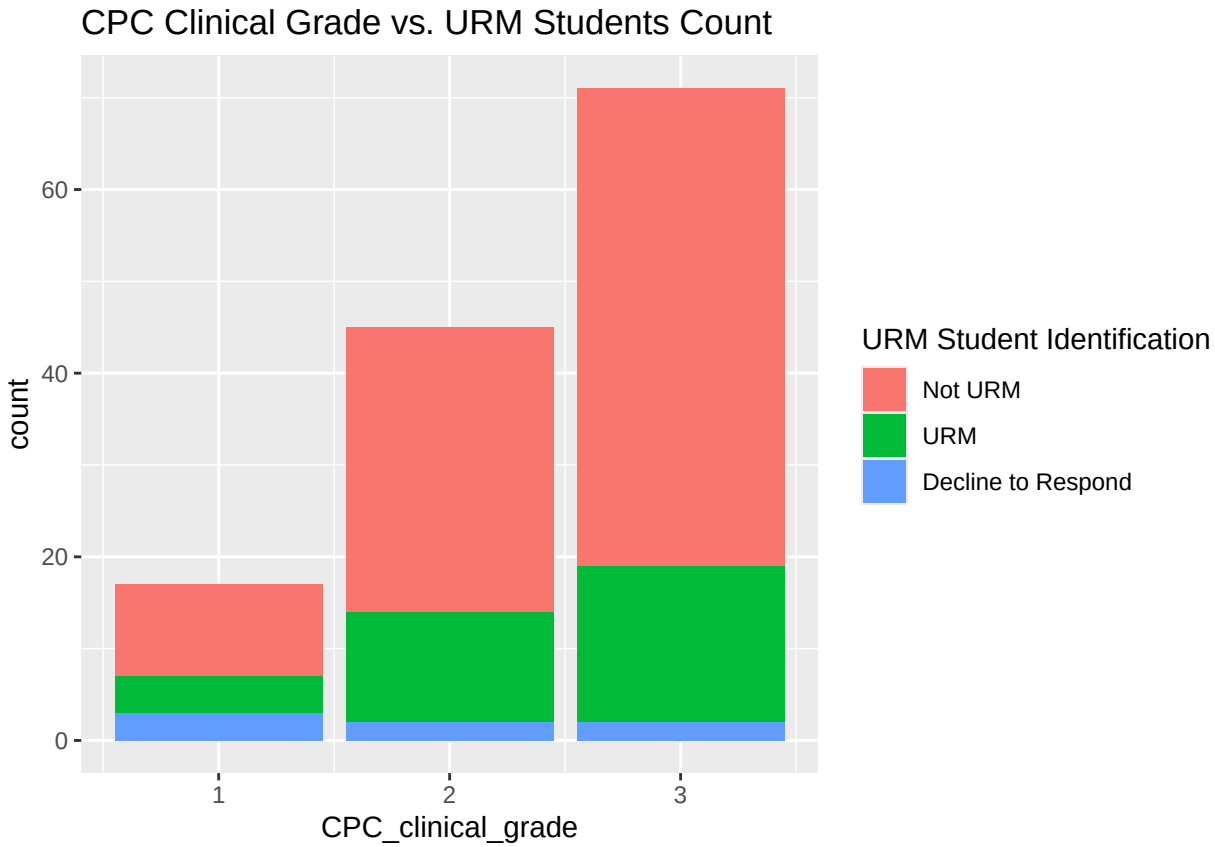
[1]Median (Q1, Q3); n (%)

```
## Warning: Removed 180 rows containing non-finite outside the scale range
## (`stat_count()`).
```

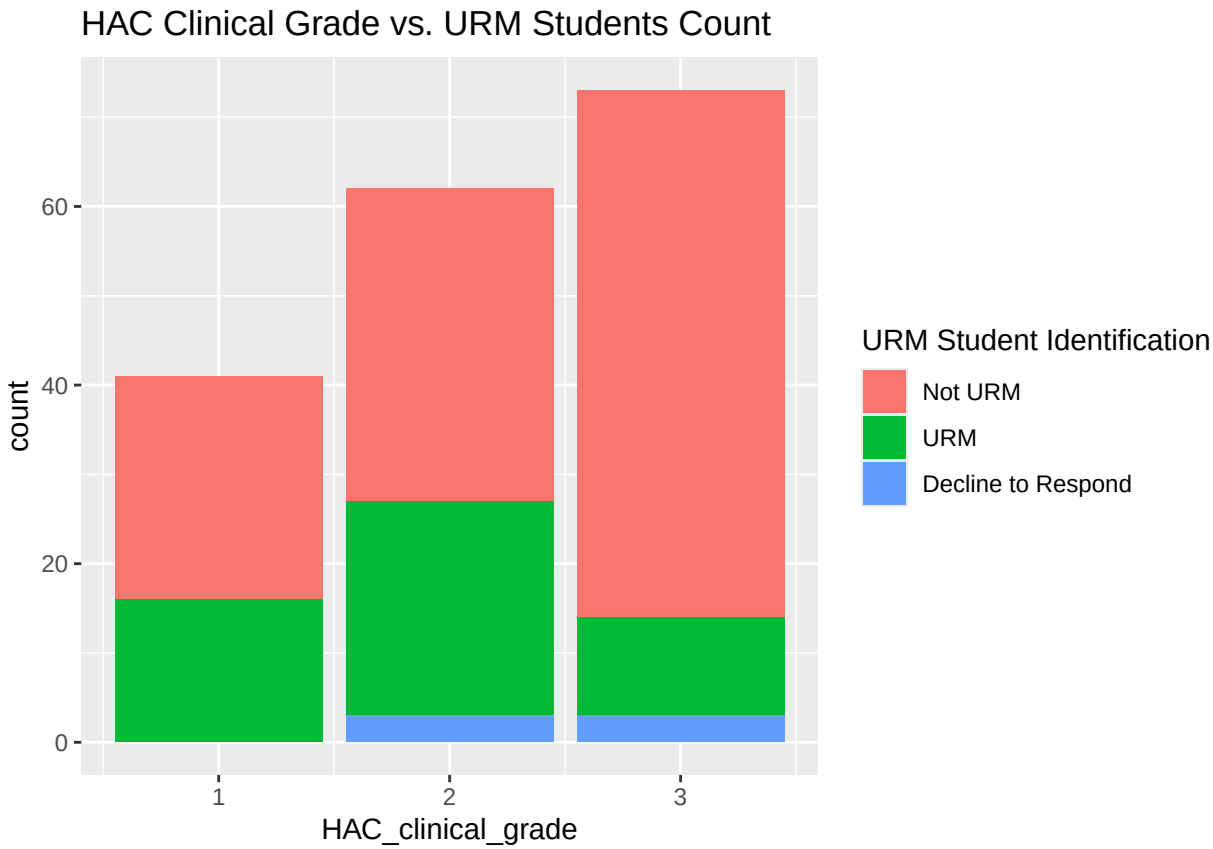



```
## Warning: Removed 66 rows containing non-finite outside the scale range
```

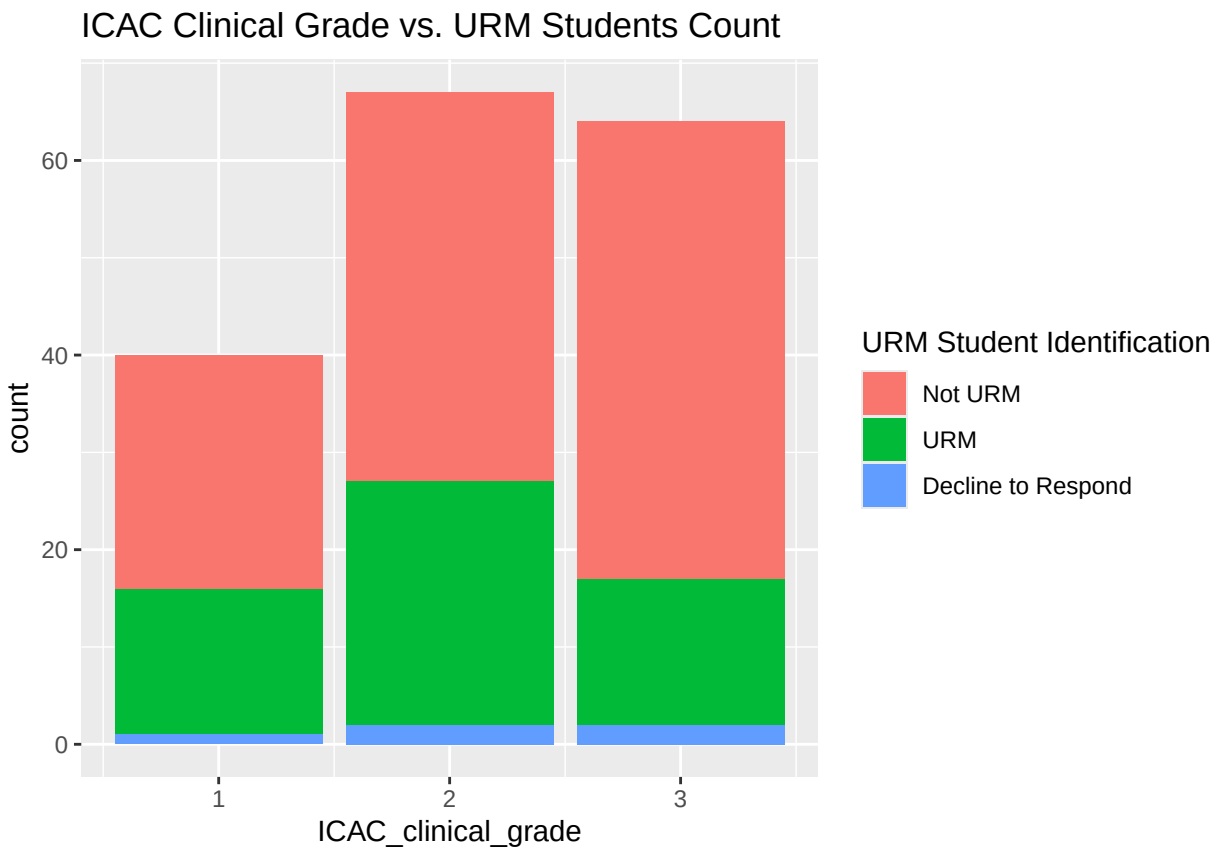
```
## (‘stat_count()‘).
```



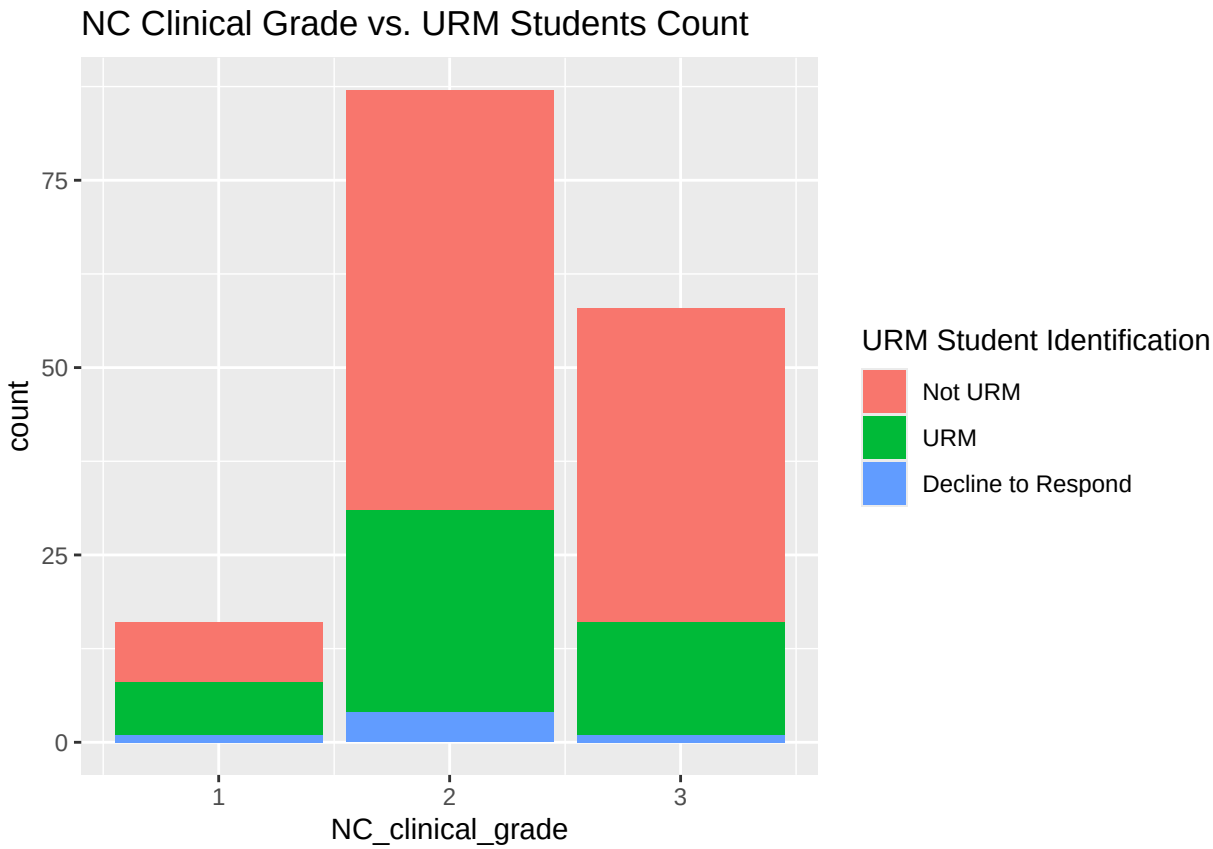

```
## Warning: Removed 23 rows containing non-finite outside the scale range
## (‘stat_count()‘).
```



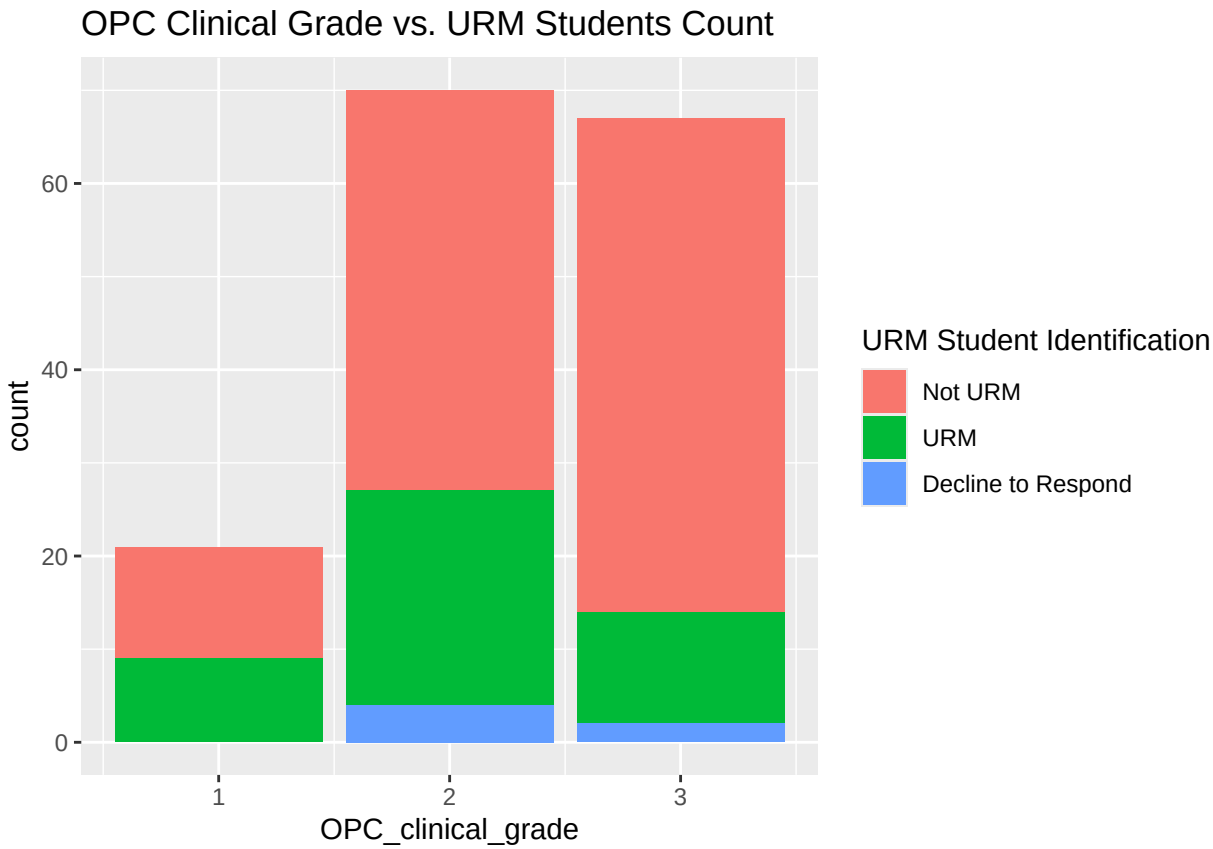

```
## Warning: Removed 28 rows containing non-finite outside the scale range
## (`stat_count()`).
```



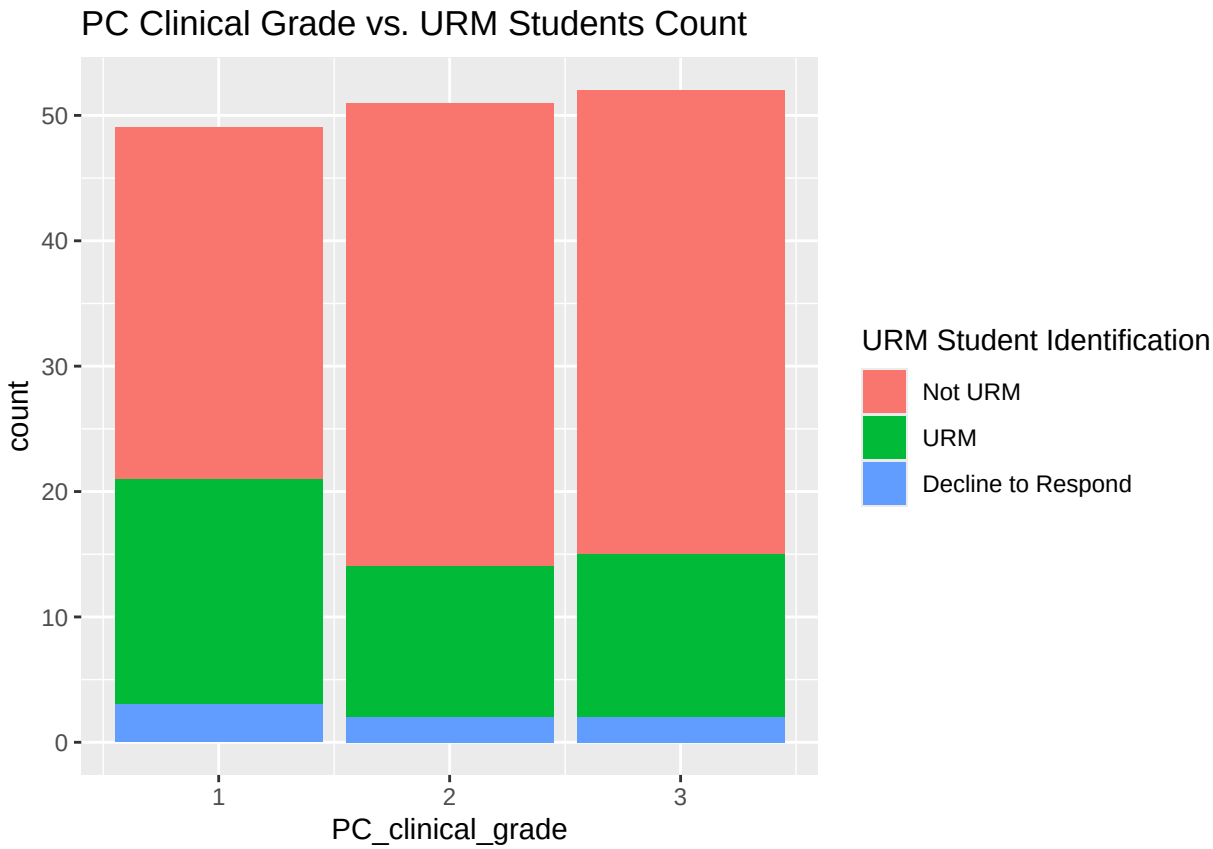

```
## Warning: Removed 38 rows containing non-finite outside the scale range
## (`stat_count()`).
```



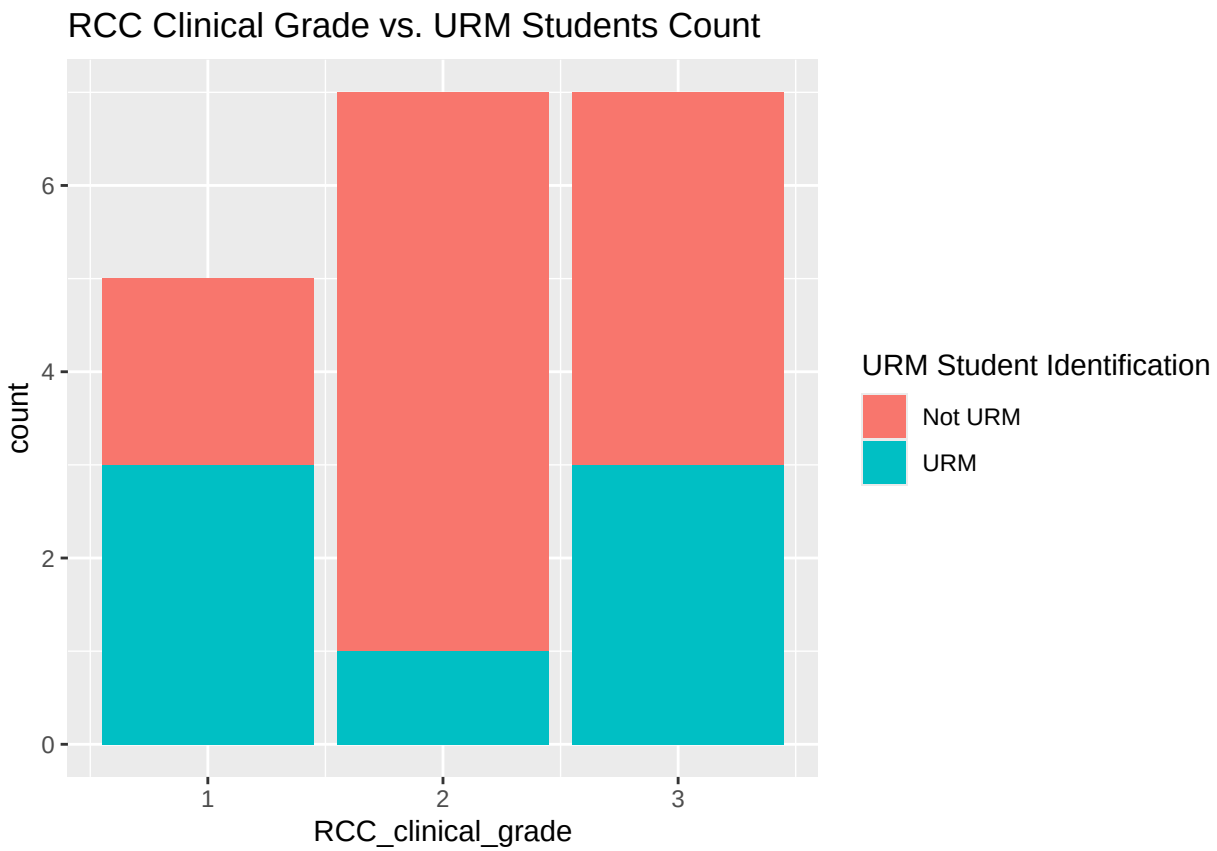

```
## Warning: Removed 41 rows containing non-finite outside the scale range
## (`stat_count()`).
```




```
## Warning: Removed 47 rows containing non-finite outside the scale range
## (`stat_count()`).
```




```
## Warning: Removed 180 rows containing non-finite outside the scale range
## (`stat_count()`).
```



Code Appendix

```
library(gtsummary)
library(tidyverse)
library(glue)
library(flextable)
set_flextable_defaults(fonts_ignore = TRUE)
# ================================================================================ loads
# in dataframe
df <- read.csv(here::here("DataRaw", "BIOS6621_Guth_data.csv"))
# View(df)
# ================================================================================ checks
# for which columns have NA values
na_df <- colSums(is.na(df))
# na_df

# confirms that the AAC_clinical_grade values and the RCC_clinical_grade values are not
# missing at random. is.na(df$AAC_clinical_grade) == is.na(df$RCC_clinical_grade)
# ================================================================================ creates
# new dataframes for the calculation of the physical exam scores and the communication
# scores for the OSCE exam respectively
phys_df <- df %>%
    select(ends_with("PhysExamScore"))
# colnames(phys_df)
```

```r
comm_df <- df %>%
    select(ends_with("CommScore"))
# colnames(comm_df)

phys_df <- phys_df %>%
    mutate(OSCE_phys_exam_avg = (P1SprBates_PhysExamScore + P1FallBodySys_PhysExamScore + P2FallRamos_Ph
        P2FallPayne_PhysExamScore + P2SpringHyde_PhysExamScore + P2SpringSnelling_PhysExamScore)/(ncol(p

comm_df <- comm_df %>%
    mutate(OSCE_comm_exam_avg = (P1SprBates_CommScore + P2FallRamos_CommScore + P2FallPayne_CommScore +
        P2SpringHyde_CommScore + P2SpringSnelling_CommScore)/(ncol(comm_df)))
# ================================================================================
final_df <- df %>%
    select(ends_with("clinical_grade"), URM, STEP1_ExamScore, STEP2CK_ExamScore, gender)

final_df$OSCE_phys_exam_avg <- phys_df$OSCE_phys_exam_avg

final_df$OSCE_comm_exam_avg <- comm_df$OSCE_comm_exam_avg

# compares those missing OSCE scores to those not missing OSCE scores
missing_OSCE <- final_df %>%
    filter(is.na(OSCE_comm_exam_avg))
non_missing_OSCE <- final_df %>%
    filter(!is.na(OSCE_comm_exam_avg))

# View(missing_OSCE) View(non_missing_OSCE)
final_df %>%
    tbl_summary(by = URM) %>%
    modify_caption("Data stratified by URM Status") %>%
    as_flex_table() %>%
    set_table_properties("autofit")

# tbl_summary(final_df, by = gender)
# ================================================================================
ggplot(data = final_df, aes(x = OBG_clinical_grade, fill = factor(URM))) + geom_bar() + labs(title = "OB
    scale_fill_discrete(name = "URM Student Identification", labels = c("Not URM", "URM", "Decline to Re
# ================================================================================
par(mfrow = c(1, 2))
ggplot(data = final_df, aes(x = factor(URM), y = STEP1_ExamScore)) + geom_boxplot() + labs(title = "STEP
ggplot(data = final_df, aes(x = factor(URM), y = STEP2CK_ExamScore)) + geom_boxplot() + labs(title = "ST
# ================================================================================
par(mfrow = c(1, 2))
ggplot(data = final_df, aes(x = factor(URM), y = OSCE_phys_exam_avg)) + geom_boxplot() + labs(title = "A
ggplot(data = final_df, aes(x = factor(URM), y = OSCE_comm_exam_avg)) + geom_boxplot() + labs(title = "A
# ================================================================================
missing_OSCE %>%
    tbl_summary() %>%
    modify_caption("Missing OSCE") %>%
    as_flex_table() %>%
    set_table_properties("autofit")
# ================================================================================
non_missing_OSCE %>%
    tbl_summary() %>%
```

```
    modify_caption("**Table 3. Non-Missing OSCE**") %>%
    as_flex_table() %>%
    set_table_properties("autofit")
# ================================================================================
ggplot(data = final_df, aes(x = AAC_clinical_grade, fill = factor(URM))) + geom_bar() + labs(title = "A
    scale_fill_discrete(name = "URM Student Identification", labels = c("Not URM", "URM", "Decline to R

ggplot(data = final_df, aes(x = CPC_clinical_grade, fill = factor(URM))) + geom_bar() + labs(title = "C
    scale_fill_discrete(name = "URM Student Identification", labels = c("Not URM", "URM", "Decline to R

ggplot(data = final_df, aes(x = HAC_clinical_grade, fill = factor(URM))) + geom_bar() + labs(title = "H
    scale_fill_discrete(name = "URM Student Identification", labels = c("Not URM", "URM", "Decline to R

ggplot(data = final_df, aes(x = ICAC_clinical_grade, fill = factor(URM))) + geom_bar() + labs(title = "
    scale_fill_discrete(name = "URM Student Identification", labels = c("Not URM", "URM", "Decline to R

ggplot(data = final_df, aes(x = NC_clinical_grade, fill = factor(URM))) + geom_bar() + labs(title = "NC
    scale_fill_discrete(name = "URM Student Identification", labels = c("Not URM", "URM", "Decline to R

ggplot(data = final_df, aes(x = OPC_clinical_grade, fill = factor(URM))) + geom_bar() + labs(title = "O
    scale_fill_discrete(name = "URM Student Identification", labels = c("Not URM", "URM", "Decline to R

ggplot(data = final_df, aes(x = PC_clinical_grade, fill = factor(URM))) + geom_bar() + labs(title = "PC
    scale_fill_discrete(name = "URM Student Identification", labels = c("Not URM", "URM", "Decline to R

ggplot(data = final_df, aes(x = RCC_clinical_grade, fill = factor(URM))) + geom_bar() + labs(title = "R
    scale_fill_discrete(name = "URM Student Identification", labels = c("Not URM", "URM", "Decline to R
```